

ABHAY RAJ GURRAM

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MACHINE LEARNING / AI ENGINEERING INTERN

Machine Learning Engineer specializing in applied AI systems, large language models (LLMs), reinforcement learning concepts, and intelligent agent-based workflows. Strong background in Python and PyTorch with hands-on experience developing, fine-tuning, evaluating, and integrating ML models into production-style software environments.

TECHNICAL SKILLS

- Programming: Python (Advanced), SQL
- ML Frameworks: PyTorch, TensorFlow, Scikit-learn
- LLM & AI Tools: Hugging Face Transformers, LangChain, AutoGen, RAG pipelines
- Reinforcement Learning: Policy gradients (conceptual), value functions, reward modeling
- Data Engineering: Pandas, NumPy, synthetic data generation, preprocessing pipelines
- Dev Tools: Git, Docker (basic), REST APIs, AWS fundamentals

EXPERIENCE

Quality Assurance Intern – Laurus Labs Ltd. (Jan 2022 – June 2022)

- Worked with structured datasets and automated validation workflows to ensure data integrity.
- Performed root cause analysis on data inconsistencies and optimized validation processes.
- Collaborated with cross-functional engineering and compliance teams on data-driven workflows.

PROJECT EXPERIENCE

- **LLM-Based Agentic AI System** – Designed and implemented agent-based AI workflows using LangChain and transformer models.
- Integrated retrieval-augmented generation (RAG) pipelines for contextual accuracy.
- Fine-tuned transformer models on domain datasets and built evaluation benchmarks.
- Optimized prompts and implemented structured evaluation metrics.
- **Reinforcement Learning Exploration** – Implemented policy gradient-based RL models in simulated environments.
- Analyzed reward functions and policy stability through controlled experimentation.
- Documented results with structured logs and performance comparisons.
- **Synthetic Data & Model Training** – Generated synthetic datasets to address class imbalance.
- Designed reproducible training pipelines and evaluated model performance metrics.

EDUCATION

Master's in Business Intelligence & Analytics – Stevens Institute of Technology

Relevant Coursework: Machine Learning, Deep Learning, Optimization, Data Engineering, Statistical Modeling